WORK CARRIED OUT

The development of the Multipurpose Agricultural Robot aimed to integrate multiple agricultural processes—such as digging, seeding, levelling, irrigation—into a single automated system. The robot was designed to reduce manual labour, improve accuracy, and optimize resource usage in farming activities.

The project's success depended on the effective combination of mechanical design, electronic control, embedded programming, and wireless communication. A major part of the work carried out involved constructing a strong and stable robot frame, integrating sensors and actuators, interfacing the Arduino UNO microcontroller, and establishing reliable wireless communication through the HC-05 Bluetooth module.

The frame serves as the chassis of the robot and provides a foundation for connecting four motors—two geared DC motors for the main movement and two toy motors for auxiliary agricultural operations.

The two geared motors are connected to the rear wheels and provide the primary drive mechanism. These motors deliver high torque at low speed, making them suitable for ploughing and carrying loads. The H-Bridge motor driver (L293D) is used to control the direction of motion—forward, backward, left, and right—based on the commands received from the Arduino UNO.

The two toy motors are employed for additional agricultural operations such as:

Digging and Seeding: The digging unit prepares the soil by loosening it, while the seeding unit drops seeds at uniform intervals.

Levelling: The levelling blade ensures an even surface for irrigation and growth.

These mechanisms are controlled independently through separate channels of the motor driver circuit, programmed to execute operations sequentially or simultaneously depending on the selected mode.

The robot is designed to perform multiple agricultural tasks, each corresponding to a specific operational mode. These modes are activated through different control loops in the Arduino program.

Digging Mode: In this mode, the toy motors connected to a rotary digging blade are activated. The robot moves forward slowly while the blades rotate, tilling the soil for seeding. The speed and direction are controlled by PWM (Pulse Width Modulation) signals generated by the Arduino.

Seeding Mode: After digging, the robot enters the seeding mode. A solenoid or small DC motor releases seeds from the hopper at predetermined intervals. Timing control ensures uniform seed spacing, improving crop consistency.

Levelling Mode: The levelling mode is used to smooth the soil surface after seeding. A flat metallic blade attached to the front of the robot is driven by a toy motor to remove irregularities on the surface.

Irrigation Mode: The irrigation mode uses a water pump connected to a moisture sensor circuit. The sensor continuously monitors soil moisture and triggers the pump whenever the moisture level falls below a predefined threshold. This ensures optimum watering without human supervision.

The HC-05 Bluetooth module is a critical component that provides wireless serial communication between the agricultural robot and external devices such as smartphones or computers. It operates on the Bluetooth 2.0 + EDR (Enhanced Data Rate) standard and uses the UART (Universal Asynchronous Receiver/Transmitter) interface for data transmission.

The HC-05 enables bidirectional communication—sending data from the Arduino to the user and receiving control commands from the Bluetooth terminal application on the user's mobile device.

Hardware Interface

The HC-05 module is connected to the Arduino UNO as follows:

- VCC → 5V

- GND → GND
- TXD → RX (Pin 0)
- RXD → TX (Pin 1)
- EN (Key) – Used for AT command mode (optional).

Communication Process

1. The user connects their smartphone to the robot's HC-05 Bluetooth network.
2. Using a Bluetooth Terminal App, the user sends single-character commands such as:
    - 'F' → Move Forward
    - 'B' → Move Backward
    - 'L' → Turn Left
    - 'R' → Turn Right
    - 'S' → Stop
    - 'P' → Sprinkle
    - 'Z'→ Levelling
    - 'E'→ Sowing
    - 'D'→ Digging
    - 'T'→ Moisture Sensing
3. The Arduino reads the incoming serial data using the Serial. read() function and executes the corresponding routine.

Implementing an agricultural robot equipped with a diverse range of functionalities, including seeding, digging, mudding, water and pest sprinkler, and leaf disease detection, involves a comprehensive and iterative process that encompasses various stages, each demanding meticulous attention to detail and a deep understanding of agricultural requirements. The journey begins with hardware selection, where careful consideration is given to choosing the right components that can withstand the rigors of agricultural environments while delivering optimal performance. Actuators capable of exerting sufficient force for digging and mudding tasks are carefully selected, while sensors capable of accurately detecting soil moisture levels, pest activity, and leaf diseases are essential for effective monitoring and control. Additionally, a robust navigation system comprising GPS, inertial sensors, and odometer is necessary to enable autonomous movement and localization within the agricultural field.

In implementing an agricultural robot with seeding, digging, mudding, water and pest sprinkler, and leaf disease detection functionalities is a complex and challenging process that demands expertise in mechanical engineering, electronics, software development, and agricultural science.

By following a systematic approach encompassing hardware selection, software development, integration, testing, and iterative improvement, it is possible to create a sophisticated robotic system capable of enhancing efficiency, productivity, and sustainability in agricultural operations.

In addition to the core functionalities of seeding, digging, mudding, water and pest sprinkler, and leaf disease detection, the implementation of an agricultural robot may also incorporate features to enhance its versatility and adaptability to diverse farming practices. For instance, modular design principles can be employed to facilitate the integration of additional tools or attachments, allowing the robot to perform a wider range of tasks such as weeding, fertilizing, or crop monitoring. Moreover, connectivity features such as wireless communication and data sharing capabilities enable seamless integration with farm management systems, enabling farmers to remotely monitor and control the robot's operations, access real-time data insights, and optimize decision-making processes.

In the advanced phase of the project, both the CNN and YOLO models are ensembled to combine their strengths for superior disease detection performance. The CNN model provides high accuracy in classifying specific leaf diseases, while YOLO enables real-time localization of affected areas through bounding boxes. This ensemble approach enhances the reliability and robustness of the system, allowing it to perform both precise classification and live detection simultaneously. As a result, the integrated model ensures faster, more accurate, and practical disease identification — making it highly effective for on-field agricultural applications and precision crop management.

CONCLUSION

A frame is used as a base for connecting and assembling everything in the agricultural robot, it consists of four motors. Two of which are gear motors and other two are toy motors. The robot encompasses following functions:

1. Digging
2. Seeding
3. Levelling
4. Disease Detection
5. Irrigation

These functions are done using different modes. Programming for different modes is done using different loops and functions. The combined implementation of embedded control and Bluetooth connectivity transformed agricultural assistant, capable of performing essential farming tasks with high accuracy and reduced human effort.

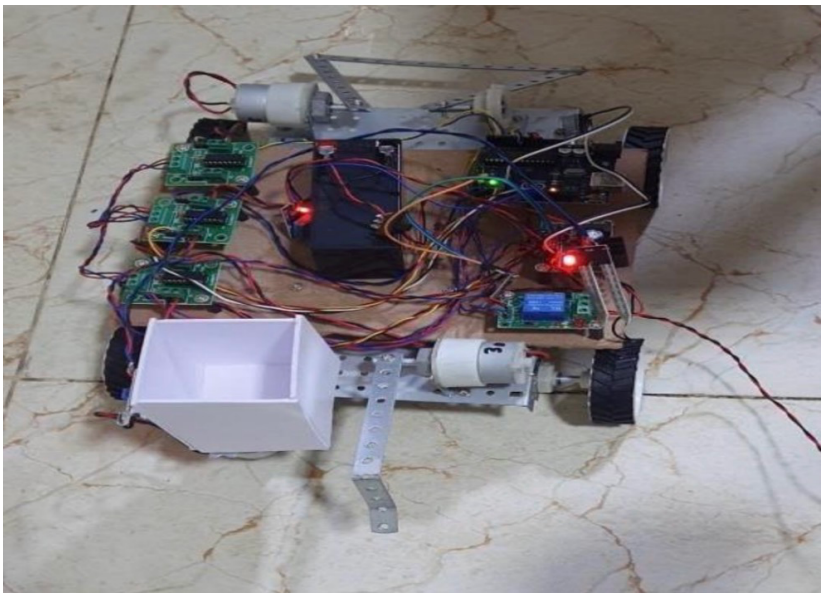


Fig.1.Working model

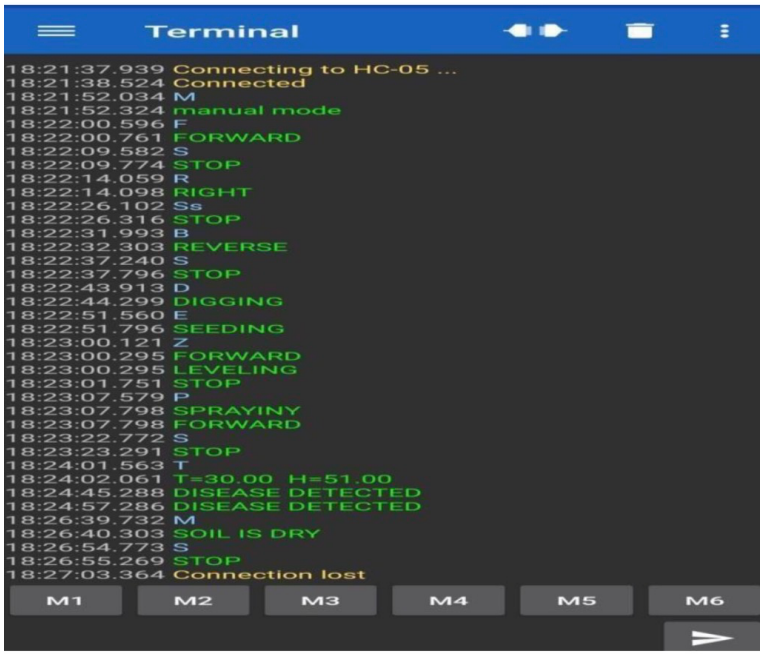


Fig.2.Working terminal

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